Kadi-APY for everyday lab work

Online workshop, participant guide. Please complete the checklist on this page before the workshop. It takes about 20 minutes.

DATE & TIME
Friday, 13 November 2026, 14:00-16:00 (CET)

Checklist before the workshop

- [ ] **1. Log in to Kadi4Mat**
  Open **kadi4mat.iam.kit.edu** and log in. Kadi4Mat is the research data repository used in the workshop.

- [ ] **2. Create a personal access token**
  The token allows kadi-apy to access Kadi4Mat on your behalf. In Kadi4Mat, click your name (top right), then Settings > Access tokens. Name it `workshop`, set an expiry date a few days after the workshop, and copy the token right away, it is only shown once.

- [ ] **3. Open the workshop in GitHub Codespaces**
  Sign in to GitHub (a free account is enough), open **codespaces.new/msishuvo/thematic-workshop-Kadi-APY** and click "Create codespace". VS Code opens in the browser with the workshop files. The first start takes a few minutes.

- [ ] **4. Install and configure kadi-apy**
  Open a terminal (Terminal > New Terminal) and run, as on your own computer:

  ```
  pip install kadi-apy matplotlib
  kadi-apy config create          # creates the configuration file ~/.kadiconfig
  kadi-apy config set-host        # enter https://kadi4mat.iam.kit.edu
  kadi-apy config set-pat         # paste your token
  kadi-apy config get-kadi-info   # checks the connection
  ```

  The token is visible while you paste it, so do not share your screen during this step.

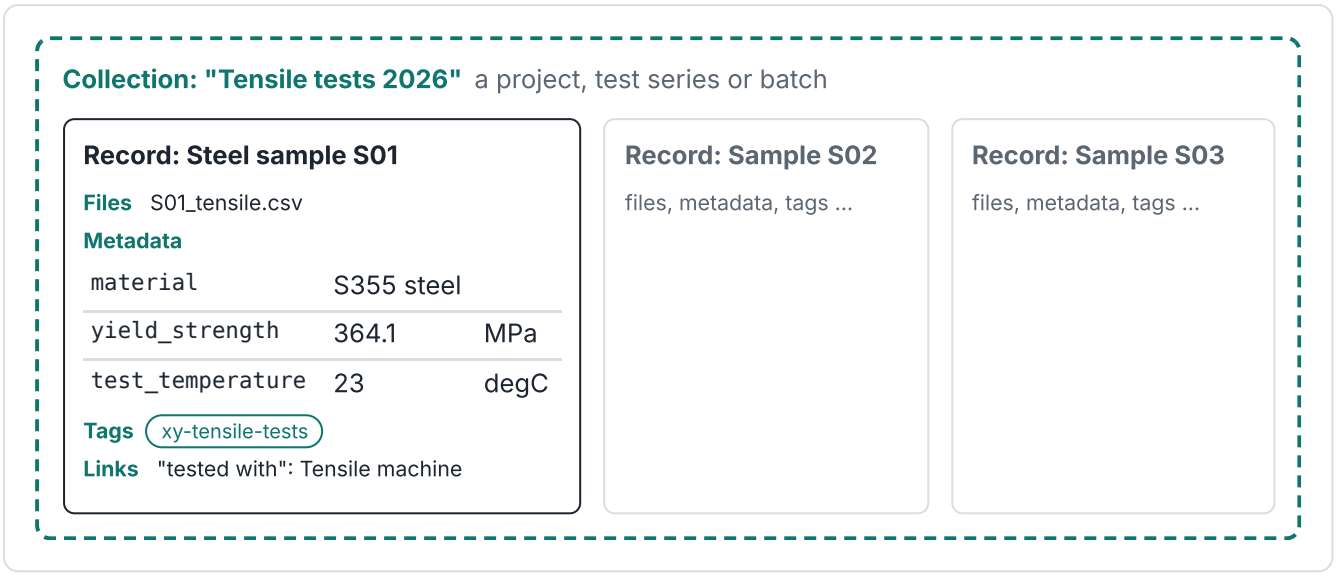
- [ ] **5. Open the notebook and run the setup cells (section 0)**
  Open `notebooks/kadi_apy_workshop.ipynb`. If VS Code asks to install the Python or Jupyter extension, click Install. Choose the kernel: "Select Kernel" (top right) > Python Environments. Run the cells with `Shift` + `Enter` and replace `"xy"` in cell 0.4 with your initials. Done when the output shows "Setup complete", "Connected to ..." and "Prefix set to ...".

- [ ] **6. Prepare your screen**
  Test your microphone and the meeting chat. If possible, use two screens or place the meeting and VS Code side by side. Keep Kadi4Mat open in a second browser tab.

> **Problems with the setup?** Please reply to the invitation e-mail with a screenshot of the error before the workshop, or join the meeting 10 minutes early. If Codespaces does not work for you, Google Colab is the backup: see "Backup: Google Colab" in `docs/participant-setup.md` in the repository.

Kadi4Mat basics

Kadi4Mat stores research data together with structured, searchable metadata. The main terms used in the workshop:



TERM	MEANING
Record	One documented object, e.g. a sample, a measurement or a simulation run. Contains files and metadata.
Metadata	Structured information about a record: key (name), type (text, number, ...), value and unit. Metadata can be searched.
Collection	A group of records, e.g. a project or a test series.
Tag	A keyword for finding records.
Identifier	A short name for a record, e.g. `jd-sample-s01`. It must be unique in Kadi4Mat, so in the workshop it starts with your initials.

What is kadi-apy?

kadi-apy is a Python library and command line tool for the Kadi4Mat API. Most actions of the web interface can be scripted with it, which is useful when the same steps are repeated for many files.

Website
Manual work, suitable for a few records.

Command line
`kadi-apy records ...`
One command per action.

Python
`from kadi_apy import ...`
Loops, calculations, reading files.

> **Typical uses:** uploading a folder of instrument files; extracting results (e.g. yield strength) into metadata; consistent names and tags within a group; downloading data for a report or paper.

Working in VS Code

TASK	HOW
Open a terminal	Terminal > New Terminal (or `Ctrl` + `` ` ``).
Run a cell	`Shift` + `Enter`, or the run button on the left of the cell.
Order	From top to bottom. Later cells use results of earlier ones.
Structure	Each part: example, reference with documentation links, exercise.
Exercises	Change only lines marked `TODO`. Solution: "Show solution".
Lines starting with `!`	Terminal commands in a notebook; in the terminal without `!`.
Reset	"Restart" in the notebook toolbar, then run again from the top.
Errors	The last line of the error message usually names the problem.

Using the command line (CLI)

All kadi-apy commands have the same structure:

`kadi-apy` the program	**`records`** resource type: records, collections, ...	**`add-metadatum`** the action	**`-R jd-sample-s01 -m length -v 80 -t float -u mm`** options, each as **-letter value**

Where to run commands

Terminal (VS Code, Codespaces, own computer):

```
kadi-apy records --help
```

Notebook cell: with `!` in front:

```
!kadi-apy records --help
```

Notes

Values with spaces need quotes: `-t "Steel sample S01"`.
Options are case-sensitive: `-R` and `-r` are different.
`--help` after any command lists its options.

Options used in the workshop

OPTION	MEANING
`-i`	identifier of a **new** record (`create`)
`-R`	identifier of an **existing** record
`-t`	title (`create`), tag (`add-tag`) or type (`add-metadatum`)
`-n`	file or folder to upload
`-m -v`	metadata name and value
`-u`	unit, e.g. `mm` , `MPa`
`-f`	overwrite existing files or metadata
`--help`	show all options of a command

During the workshop
Please write in the chat when you have finished an exercise, and post questions or error messages there at any time.

After the workshop
Documentation: kadi-apy.readthedocs.io
Material: github.com/msishuvo/thematic-workshop-Kadi-APY
Unpublished lab data: check your data policy before using cloud services.

Command reference

Full documentation: **kadi-apy.readthedocs.io**, pages Usage > Library (Python) and Usage > CLI. In the notebook: `help(Record.upload_file)`; in the terminal: `kadi-apy records add-files --help`.

Command line (in the terminal; in a notebook cell, put `!` in front)

```
# create a record
kadi-apy records create -i jd-sample-s01 -t "Sample S01"
# upload a file (or a whole folder)
kadi-apy records add-files -R jd-sample-s01 -n data/tensile_tests/S01_tensile.csv
# add metadata with type and unit
kadi-apy records add-metadatum -R jd-sample-s01 -m specimen_length -v 80 -t float -u mm
# add a tag
kadi-apy records add-tag -R jd-sample-s01 -t jd-tensile-tests
# show metadata (-m) and file list (-l)
kadi-apy records show-info -R jd-sample-s01 -m -l
# find records by tag / list all commands
kadi-apy records get-records -t jd-tensile-tests
kadi-apy records --help
```

Python

```
from kadi_apy import KadiManager
manager = KadiManager()                        # connection, reads the config file

# get the record, or create it if it does not exist yet
record = manager.record(identifier="jd-sample-s01", title="Sample S01", create=True)

# metadata: a list of dictionaries (key, type, value, unit)
record.add_metadata([{"key": "specimen_length", "type": "float",
                      "value": 80.0, "unit": "mm"}], force=True)

record.upload_file("data/tensile_tests/S01_tensile.csv", force=True)   # upload a file
record.add_tag("jd-tensile-tests")                    # add a tag

# put the record into a collection
collection = manager.collection(identifier="jd-tensile-tests-2026", title="Tensile tests 2026",
create=True)
record.add_collection_link(collection.id)
```

Common errors

MESSAGE	CAUSE AND SOLUTION
NameError: name 'manager' is not defined	An earlier cell was not run, e.g. after a restart. Run section 0 again.
Config file does not exist	Run `kadi-apy config create`, `set-host` and `set-pat` in the terminal.
No module named 'kadi_apy'	Wrong kernel, or kadi-apy not installed for it. Select the kernel again or run `%pip install kadi-apy` in a cell.
401 / "invalid access token"	Token incomplete or expired. Create a new token and run `kadi-apy config set-pat` again.
identifier ... already in use	The identifier belongs to someone else. Check your prefix.
SyntaxError, IndentationError	Missing quote, bracket or comma, or wrong indentation. Compare with the solution.